



**SERIAL PROTOCOL FOR THE μ MAC
INSTRUMENTS
Ver. 2.1.4**

**Valid only for μ MAC with firmware Ver.
2.5.11 and over**

Revised: 15 February 2007

Serial port settings: 9600,N,8,1

DESCRIPTION	COMMAND	REPLY MESSAGE
Start a complete analytical cycle on the active methods	#C<CR><LF>	<ACK><CR><LF> or <NAK><CR><LF> or BUSY<CR><LF>
Start a complete calibration cycle on the active methods	#K<CR><LF>	<ACK><CR><LF> or <NAK><CR><LF> or BUSY<CR><LF>
Start a complete blank cycle on the active methods	#Y<CR><LF>	<ACK><CR><LF> or <NAK><CR><LF> or BUSY<CR><LF>
Start a wash cycle on the instrument	#W<CR><LF>	<ACK><CR><LF> or <NAK><CR><LF> or BUSY<CR><LF>
Stop instrument Attention: Using this command during an analytical cycle you have to run a wash cycle (#W) before to use again the instrument	#X<CR><LF>	<ACK><CR><LF> or <NAK><CR><LF> or BUSY<CR><LF>
Request of last result with O.D. Attention: These values will be updated at the end of each analysis cycle or calibration cycle • Fixed length answer POSSIBLE ERRORS CODE : 22222 ->Missing sample 1 33333 ->Missing sample 2 55555->Leakage error 77777->Out of range 88888->Air in cell	#D0<CR><LF>	XXX<SP>Date_and_hour<CR><LF>AAAAA.AA<SP> AAAAA.AA<SP>AAAAA.AA<SP>AAAAA.AA<CR><LF> >BBBBB.BB<SP>BBBBB.BB<SP>BBBBB.BB<SP>BB BBB.BB<CR><LF>CCCCC.CC<SP>CCCCC.CC<SP>CC CCC.CC<SP>CCCCC.CC<CR><LF> or <NAK><CR><LF> Where: • "XXX" can be • 000=Analysis cycles results • 100=Calibration cycles results • Date_and_hour → 16 chars • "AAAAA.AA" • if XXX=000 AAAAA.AA=conc. results • if XXX=100 AAAAA.AA=calibration O.D. • For errors code please see on the left • "CCCCC.CC" → final O.D. Note: O.D. = Optical Density Example: 000 MM/DD/YY HH:MM 1.500000 2.200000 2.560000 0.730000 0.043000 0.092000 0.062000 0.016000 0.520000 0.486000 0.561000 0.233000
Request system status	#S<CR><LF>	STAND-BY<CR><LF> or BUSY<CR><LF>
Set active methods	#Pxxxx<CR><LF> Where x is: ▪ 0 if not active ▪ 1 if active	<ACK><CR><LF> or <NAK><CR><LF> or BUSY<CR><LF>

DESCRIPTION	COMMAND	REPLY MESSAGE
Read the active methods	#J<CR><LF>	XXXX<CR><LF> Where X is: ▪ 0 if not active ▪ 1 if active
Reset internal results counter index	#H<CR><LF>	<ACK><CR><LF> or <NAK><CR><LF> or BUSY<CR><LF>
Start download of the selected results <u>RS485 and RS232 Interface</u> • Not fixed length answer (depend on method's name)	#USSSEEE<CR><LF> SSS = Start position EEE = End position Position= from 000 to 400	Start the ASCII upload of the selected results buffer using RS485/RS232 interface. Single record format: Date_and_hour;METH;AAAAA.AA;BBBBB.BB;CCCCC.CC<CR><LF> Where: • Date_and_hour → 16 chars • "METH" → method's name (variable length) • "AAAAA.AA" → concentration results • "BBBBB.BB" → start O.D. • "CCCCC.CC" → final O.D. This is an example of one single record format: MM/DD/YY HH:MM;NO3;1.500000;0.021300;0.305000
Start a SEQ5 cycle on method 0. Normally used for REAGENTS PRIME.	#R<CR><LF>	<ACK><CR><LF> or <NAK><CR><LF> or BUSY<CR><LF>
Set new calibrant and blank O.D. for all methods.	#BK1KKKY1YYYK2KKKY2YYYK3KKKY3YYYK4KKKY4YYY<CR><LF>	<ACK><CR><LF> or <NAK><CR><LF> Where: • K1KKK → New calibrant O.D. for meth. 1 • Y1YYY → New blank O.D. for meth. 1 • K2KKK → New calibrant O.D. for meth. 2 • Y2YYY → New blank O.D. for meth. 2 • K3KKK → New calibrant O.D. for meth. 3 • Y3YYY → New blank O.D. for meth. 3 • K4KKK → New calibrant O.D. for meth. 4 • Y4YYY → New blank O.D. for meth. 4
Set new calibrant concentration and full scale for all methods.	#FK1KKKF1FFFK2KKKF2FFFK3KKKF3FFFK4KKKF4FFF<CR><LF>	<ACK><CR><LF> or <NAK><CR><LF> Where: • K1KKK → New calibrant conc. for meth. 1 • F1FFF → New full scale value for meth. 1 • K2KKK → New calibrant conc. for meth. 2 • F2FFF → New full scale value for meth. 2 • K3KKK → New calibrant conc. for meth. 3 • F3FFF → New full scale value for meth. 3 • K4KKK → New calibrant conc. for meth. 4 • F4FFF → New full scale value for meth. 4

DESCRIPTION	COMMAND	REPLY MESSAGE
Request analytical system info Fixed length answer <u>See also, on next page, the new extend command #g (lowercase)</u>	#G<CR><LF>	AAAAA.AA<SP>BBBBB.BB<SP>C<SP>D<SP>E<SP>F<SP>GGGG<CR><LF> or <NAK><CR><LF> Where: - AAAAA.AA → Real time O.D. - BBBBB.BB → Real time temperature - C → Stand-by or Busy (0-1) - D → Active method (0-3) - E → Kind of analysis : 0=SAMPLE 1=CALIBRATION 2=BLANK 4=WASH 5=PRIME - F → MONITOR: NO/YES (0-1) - GGGG → MONITOR TIME NEXT ANALYSIS Example: 0.455000 25.35000 1 0 1 1 0064 Where: O.D.=0.455 Temp °C=25.35 System busy, calibration on method 0,monitor active, 964 min. for next analysis
Request global system info Fixed length answer	#A<CR><LF>	K1KKK.KK2KKK.KK3KKK.KK4KKK.KKF1FFF.FFF2FFF.FFF3FFF.FFF4FFF.FF1CCC.CCC2CCC.CCC3CCC.CC4CCC.CCY1YYY.YY2YY.YY3YYY.YY4YYY.YYXXXX<CR><LF> or <NAK> Where: - K1-K4 → Stored calibrant conc. Meth. N - F1-F4 → Stored full scale Meth. N - C1-C4 → Stored calibrant O.D. Meth. N - Y1-Y4 → Stored blank O.D. Meth. N - XXXX → Active methods: - Where X is: 0 if not active 1 if active
Read the methods' names Not fixed length answer (depend on method's name)	#L<CR><LF>	METH1<SP>METH2<SP>METH3<SP>METH4<CR><LF> Example: NH3 PO4 NH2 NH3

DESCRIPTION	COMMAND	REPLY MESSAGE
Read monitor' parameters • Fixed length answer	#M<CR><LF>	NNNNKKKH0H1H2H3H4H5H6H7H8H9<CR><LF> Where: • NNNN → Sampling frequency (minutes) • KKK→ Recalibration frequency (samples) • H0 to H9→ Recalibration hours (up to 10)
Set monitor' parameters	#NNNNKKKH0H1H2H3H4H5H6H7H8H9<CR><LF> Where: • NNNN → Sampling frequency (minutes) • KKK→ Recalibration frequency (samples) • H0 to H9 → Recalibration hours (up to 10). Allowed values 0-23. The value 24 is used as OFF.	<ACK><CR><LF> or <NAK><CR><LF> or BUSY<CR><LF>
Active and start system monitor functionality	#V<CR><LF>	<ACK><CR><LF> or <NAK><CR><LF> When monitor functionality is started the system will perform immediately a complete analytical cycle before enter in the wait mode.
Request extended analytical system info • Fixed length answer	#g<CR><LF>	AAAAA.AA<SP>BBBBB.BB<SP>C<SP>D<SP>E<SP>F<SP>GGG<SP>H<SP>JJJJ<CR><LF> or <NAK><CR><LF> Where: • AAAAA.AA → Real time O.D. • BBBBB.BB → Real time temperature • C → Stand-by or Busy (0-1) • D → Active method (0-3) • E → Kind of analysis : • 0=SAMPLE • 1=CALIBRATION • 2=BLANK • 4=WASH • 5=PRIME • F →DILUTION : NO/YES (0-1) • GGG →Step in execution • H → MONITOR: NO/YES (0-1) • JJJJ → MONITOR TIME NEXT ANALYSIS Example: 0.455000 25.35000 1 0 1 0 067 1 0094 Where: O.D.=0.455 Temp °C=25.35 System busy, calibration on method 0, No dilution, step 67, Monitor active, 94 min. for next analysis
Set system date and time	#dDDMMYYYYhhmm<CR><LF> Where: DDMMYYYY → Date hhmm → Hour	<ACK><CR><LF> or <NAK><CR><LF> or <BUSY><CR><LF>

DESCRIPTION	COMMAND	REPLY MESSAGE
Read system date and time • Fixed length answer	#t<CR><LF>	DDMMYYhhmm<CR><LF> Where: DDMMYY → Date hhmm → Hour
Read instrument serial number • Not Fixed length answer	#s<CR><LF>	SN=xxxxxxxxxx<CR><LF>